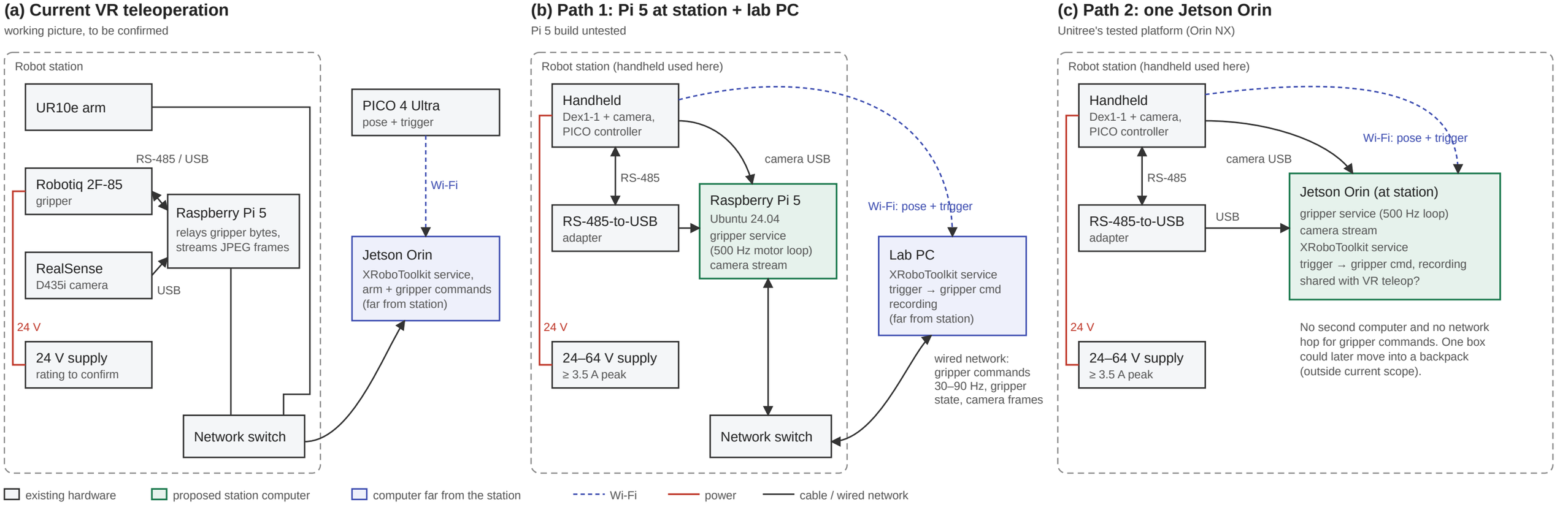
(a) Current VR teleoperation
working picture, to be confirmed
Robot station
UR10e arm
RS-485 / USB
Robotiq 2F-85
gripper
Raspberry Pi 5
relays gripper bytes,
streams JPEG frames
RealSense
D435i camera
USB
24 V
24 V supply
rating to confirm
Network switch
PICO 4 Ultra
pose + trigger
Wi-Fi
Jetson Orin
XRoboToolkit service,
arm + gripper commands
(far from station)
(b) Path 1: Pi 5 at station + lab PC
Pi 5 build untested
Robot station (handheld used here)
Handheld
Dex1-1 + camera,
PICO controller
RS-485
RS-485-to-USB
adapter
camera USB
Raspberry Pi 5
Ubuntu 24.04
gripper service
(500 Hz motor loop)
camera stream
24 V
24–64 V supply
≥ 3.5 A peak
Network switch
Wi-Fi: pose + trigger
Lab PC
XRoboToolkit service
trigger → gripper cmd
recording
(far from station)
wired network:
gripper commands
30–90 Hz, gripper
state, camera frames
(c) Path 2: one Jetson Orin
Unitree's tested platform (Orin NX)
Robot station (handheld used here)
Handheld
Dex1-1 + camera,
PICO controller
RS-485
RS-485-to-USB
adapter
camera USB
Wi-Fi: pose + trigger
USB
Jetson Orin (at station)
gripper service (500 Hz loop)
camera stream
XRoboToolkit service
trigger → gripper cmd, recording
shared with VR teleop?
24 V
24–64 V supply
≥ 3.5 A peak
No second computer and no network
hop for gripper commands. One box
could later move into a backpack
(outside current scope).
existing hardware
proposed station computer
computer far from the station
Wi-Fi
power
cable / wired network